

CS-412 Operating Systems

Assignment

Date: April 15, 2025

Marks: 20 (5*4)

Please Note.

Originality will be the most important criteria for submission of assignment. The assignment will be checked by the standard plagiarism tool of JNU and if found more than 15 % similarity, it will be rejected. Evaluation will be based on proper understanding of concepts.

Answers should be submitted in a single file in pdf format (either hand-written or computer typed). The assignment is supposed to be submitted by April 28, 2025 (Hard deadline) at (ratnesh@jnu.ac.in) with Name and course name in Subject line)

THERE IS NO PAGE LIMIT CONSTRAINT.

1. Write a program that simulates the implementation of three disk scheduling algorithms: First-Come-First-Served (FCFS), Shortest Seek Time First (SSTF), and SCAN. Test the program with a variety of inputs and analyse the average-seek time and throughput for each algorithm.
2. Design a custom CPU scheduling algorithm for an embedded system that handles real-time tasks. Justify your choices for the algorithm's implementation, and explain how it will handle priority inversion. Implement this scheduling algorithm in pseudocode and describe the data structures used.
3. Design and implement a solution to the producer-consumer problem using semaphores or mutex. Simulate a multi-threaded environment where multiple producers and consumers interact with a shared buffer, and ensure that your solution avoids deadlocks and race conditions.
4. Design a security model for an operating system that manages user access control and prevents unauthorized access to system resources. Discuss different authentication techniques, encryption methods, and access control policies. Provide pseudocode or diagrams that explain your approach.